

ML Notes

XGBoost (Extreme Gradient Boosting)

A machine learning library designed for classification and regression which will be useful in our case. High speed and performance. Key features are gradient boosting, which builds many decision trees sequentially. Each tree corrects the errors of the previous tree. It also learns how to handle missing data which could be beneficial if our dataset is incomplete. Finally it has many regularization techniques which helps us prevent overfitting, which is common on smaller datasets.

Hugging face

Provides access to pretrained models. These models are trained on large datasets which can be fine tuned to our data. This could be useful if we find one that is trained on similar data to ours. There is also an API for training models from scratch which could simplify the process of writing the software to do so. This would be an alternative to writing everything from scratch using something like pytorch, which would take a lot of time.

SK Learn

Sk learn provides easy API access to core machine learning algorithms and benchmarking/testing features. If we find an architecture we want, sk learn can greatly simplify the process of writing the code.

Literature analysis

The paper "**A statistical perspective for predicting the strength of metals: Revisiting the Hall–Petch relationship using machine learning**" presents a novel machine learning-based framework to statistically analyze and predict the flow stress (strength) of polycrystalline metals based on microstructural features, particularly focusing on grain size.

Key Points:

- **Objective:** Improve understanding of how microstructural features, especially grain size, influence metal strength. The authors aim to address inconsistencies in the traditional Hall–Petch relationship using statistical and machine learning methods.
- **Background:**
 - The Hall–Petch relationship empirically links flow stress to the inverse square root of grain size:
 - There is significant variability in reported values of parameters σ_0 , k_0 , and n across literature due to data scarcity and microstructural variability.
- **Methodology:**
 - Over 1.1 million synthetic polycrystalline FCC nickel samples were generated using physics-informed models considering grain size, orientation, and dislocation density.
 - Flow stresses were calculated for each sample at various strain levels.

- Gaussian mixture models and neural networks (specifically Mixture Density Networks) were used to predict the distribution of flow stress values probabilistically.
- **Results:**
 - The Hall–Petch relationship holds statistically for average grain sizes above a transition threshold which increases with strain.
 - For smaller grains or at high strains, deviations occur due to dominance of other deformation mechanisms (like dislocation nucleation).
 - Grain size and strain were the most significant microstructural predictors of strength.
 - The probabilistic approach captures the wide variability in flow stress that deterministic models cannot.
- **Significance:**
 - This framework enables accurate strength prediction despite the randomness in microstructure.
 - It shows that large synthetic datasets can compensate for sparse experimental data, providing deeper insights into deformation mechanisms.
 - It emphasizes the necessity of probabilistic modeling in materials science for robust design and discovery.

Simplified explanation:

In this study, the researchers used machine learning to improve how we predict the strength of metals based on their internal structure. Instead of relying only on traditional equations, they generated over a million virtual metal samples—each with different grain sizes, orientations, and defect levels—and used physics-based models to calculate how strong each sample would be. Then, they trained a special type of machine learning model called a **Mixture Density Network (MDN)**, which can predict not just a single strength value but the full range of possible strengths for a given metal structure. This approach captures the natural randomness in metal microstructures and helps show how likely certain strength values are. The machine learning model identified grain size and strain as the most important features influencing strength and provided a much more accurate, flexible tool for designing stronger metals than older methods.